

Research Project Report

Investigation of Neural Network Vulnerabilities in Traffic Sign Recognition Models to Adversarial Attacks

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Abstract

This report focuses on analyzing the security of computer vision systems in smart cars. It explores methods of fooling recognition models through physical and digital adversarial attacks. The document provides a literature review, describes the stages of detection and classification, and presents an overview of documented testing methodologies and theoretical attack scenarios based on existing research.

1 Introduction and Problem Statement

With the development of autonomous vehicles, the reliability of road infrastructure recognition has become critically important. Errors in identifying signs (e.g., ignoring a "Stop" sign or misinterpreting a speed limit) can lead to emergency situations. The purpose of this study is to theoretically explore the mechanisms of influencing neural networks to identify their weaknesses and study potential defense methods.

2 Review of Existing Literature (Research)

During the study, an in-depth analysis of current relevant English-language research was conducted. The main focus was on papers demonstrating the feasibility of conducting attacks in the real physical world.

- **Evtimov et al. (2017) "Robust Physical-World Attacks on Deep Learning Visual Classification"**: This is a foundational paper. The authors proved that ordinary stickers on signs could completely alter the model's prediction. They proposed the RP_2 algorithm, which minimizes the impact of noise and optimizes the patch to work at various viewing angles.
- **Sitawarin et al. (2018) "DARTS: Deceiving Autonomous Cars with Toxic Signs"**: Researchers proposed a method for creating "toxic" signs that appear as ordinary objects to humans but contain hidden high-frequency patterns that cause a malfunction in the car's classifier.
- **Eykholt et al. (2018) "Physical Adversarial Examples for Object Detectors"**: Unlike classic attacks on classification, this work shows how to attack object detectors (e.g., YOLO). Using special patterns, it is possible to make the system completely fail to detect the presence of a sign in the frame.

3 Image Analysis Architecture in Autonomous Vehicles

Traffic Sign Recognition (TSR) in modern systems is a two-stage process. It is important to understand the difference between the stages, as attacks on each have different consequences.

3.1 Detection

At this stage, the model (e.g., YOLOv8) analyzes the entire image from the front camera. It determines the coordinates (Bounding Box) of all potential signs. An attack on the detector makes the object "invisible" to the algorithm.

3.2 Recognition/Classification

The detected fragment is cropped and fed into a classifier (e.g., ResNet-50). The classifier's task is to assign a class label to the object (e.g., "ID: 14 — Stop"). An attack on the classifier substitutes the meaning of the sign.

4 Analysis of Testing Methods on Models

In the reviewed literature, experiments are typically conducted using models trained on the **GTSRB** (German Traffic Sign Recognition Benchmark) dataset. Existing studies evaluate two main types of influence:

4.1 Digital Attacks (Digital Noise)

Using gradient methods such as FGSM (Fast Gradient Sign Method). The essence of the method is adding small noise $\eta = \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y))$ to the original image. Research indicates that although a model's accuracy drops to zero in a digital simulation, such attacks are easily neutralized by slight compression or image rotation, making them ineffective for real-world road conditions.

4.2 Physical Patterns (Adversarial Patches)

Creating high-contrast "patches" that are placed onto the image of the sign. Studies highlight that this method proves to be significantly more stable, as the adversarial patch is generated to be invariant to shooting conditions (lighting, angles).

5 Comparative Analysis: What Works Well and What Works Poorly

Based on the analyzed literature, key patterns of vulnerabilities can be highlighted. Below is a summary table of attack effectiveness as described by researchers.

Table 1: Comparison of Attack Types (Based on Literature Review)

Attack Method	Effectiveness (Digital)	Effectiveness (Physical)	Complexity
FGSM / PGD	~98%	< 10%	Low
Adversarial Patch	~75%	~65%	High
Optical Illusions	~60%	~40%	Very High

What works poorly: Attempts to fool the model using micro-noise invisible to the eye. Car cameras have their own filters and hardware noise that "erase" the digital attack in the physical

world.

What works well: Bright, geometric shapes (patches) that the model interprets as key features of another class.

6 Theoretical Scenario of a Practical Attack

To illustrate how these vulnerabilities are exploited, we can summarize a typical attack scenario commonly described in the literature.

- **Target:** A standard Convolutional Neural Network (CNN) trained for traffic sign classification with a high baseline accuracy (e.g., $> 95\%$).
- **Attack scenario:** Applying a physical adversarial patch (such as a black-and-white sticker pattern) to the central part of a "No Entry" sign.
- **Expected Result:** The model erroneously classifies the object as a different class, such as "Speed limit 80 km/h", with high confidence. Meanwhile, the object detector continues to see the object as a traffic sign, confirming that the attack successfully targets the classifier specifically.

7 Image Analysis and Implementation Plan

If the project proceeds to a practical phase, the following algorithmic pipeline is typically required:

1. Check the frame for the presence of a sign (Stage: Detection).
2. Extract the Region of Interest (ROI).
3. Apply transformations (rotation, brightness) to simulate physical conditions.
4. Generate a patch that minimizes the loss function for the target (incorrect) class.

8 Conclusion and Future Steps

The literature review confirms the hypothesis that deep learning in TSR tasks is highly sensitive to localized physical changes, particularly adversarial patches.

References:

1. Evtimov, I., et al. (2017). "Robust Physical-World Attacks on Deep Learning Visual Classification." arXiv preprint arXiv:1707.08945.
2. Eykholt, K., et al. (2018). "Physical Adversarial Examples for Object Detectors." USENIX Workshop on Offensive Technologies (WOOT).
3. Sitawarin, C., et al. (2018). "DARTS: Deceiving Autonomous Cars with Toxic Signs." arXiv preprint arXiv:1802.06430.